

Class Activity- Exception Handling

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COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1

```
import java.util.Scanner;
```

```
class Grade {
```

```
    public static void main(String args[]) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int marks[] = {10, 15, 20, 25, 30};
```

```
        try {
```

```
            int p = sc.nextInt();
```

```
            System.out.println("Grade: " + marks[p - 1]);
```

```
        } catch (ArrayIndexOutOfBoundsException e) {
```

```
            System.out.println("Position must be between 1 and 5");
```

```
        } catch (Exception e) {
```

```
            System.out.println("Invalid input");
```

```
        }
```

```
    }
```

```
}
```



SIMATS Online Programming Lab
JAVA PROGRAMMING

 SELVA PRABHU A K
192421224RA

QUESTIONS

Q1
ArrayIndexOutOfBoundsException
10 marks

Q2
calculator application
10 marks

Q3
student registration system
10 marks

3/3 answered

Q1 ArrayIndexOutOfBoundsException
10 marks

You are developing a student grade management system. Write a program that stores 5 student grades in an array and allows users to retrieve a grade by entering the student's position (1-5). Handle cases where the position is out of range or input is invalid

Sample Output : 15

Your Code

```

1 import java.util.Scanner;
2 class Grade{
3     public static void main(String args[]) {
4         Scanner sc = new Scanner(System.in);
5         int marks[] = {10, 15, 20, 25, 30};
6         try{
7             int p = sc.nextInt();
8
9             System.out.println("Grade: " + marks[p - 1]);
10
11         }catch (ArrayIndexOutOfBoundsException e) {
12             System.out.println("Position must be between 1 and 5");
13         }catch (Exception e) {
14             System.out.println("Invalid input");
15         }
16     }
17 }

```

Input

Output

Invalid input

2

```
import java.util.Scanner;
```

```
class Calculator {
```

```
    public static void main(String args[]) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        try {
```

```
            int a = sc.nextInt();
```

```
            int b = sc.nextInt();
```

```
            System.out.println("Result: " + (a / b));
```

```
        } catch (ArithmeticException e) {
```

```
            System.out.println("Cannot divide by zero");
```

```
        }
```

```
    }
```

```
}
```

SIMATS ENGINEERING **SIMATS Online Programming Lab** **JAVA PROGRAMMING** SELVA PRABHU A K 192421224RA

QUESTIONS

- Q1** ArrayIndexOutOfBoundsException 10 marks
- Q2** calculator application 10 marks
- Q3** student registration system 10 marks

3/3 answered

Q2 calculator application 10 marks

You are building a simple calculator application. Write a program that takes two integers as input and divides them. Handle the case where the denominator is zero.

Your Code

```
1 import java.util.Scanner;
2 class Calculator{
3     public static void main(String args[]) {
4         Scanner sc = new Scanner(System.in);
5         try{
6             int a = sc.nextInt();
7             int b = sc.nextInt();
8             System.out.println("Result: " + (a / b));
9         }
10        }catch (ArithmeticException e) {
11            System.out.println("Cannot divide by zero");
12        }
13    }
14 }
```

Input

20
8

Output

Cannot divide by zero

Run **Save**

3

```
import java.util.Scanner;
```

```
class InvalidAgeException extends Exception {
    InvalidAgeException(String msg) {
        super(msg);
    }
}
```

```
public static void main(String args[]) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    String name = sc.nextLine();
```

```
    int age = sc.nextInt();
```

```
    try {
```

```
        if (age < 5 || age > 18) {
```

```
            throw new InvalidAgeException("Invalid age");
```

```
}
```

```
System.out.println("Student Name: " + name);
```

```
System.out.println("Student Age: " + age);
```

```
System.out.println("Registration Successful");
```

```
} catch (InvalidAgeException e) {
```

```
    System.out.println(e.getMessage());
```

```
}
```

```
}
```

```
}
```

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user "SELVA PRABHU A K" with ID "192421224RA". The main task is "register multiple students and display their information." The "Your Code" section contains the following Java code:

```
1 import java.util.Scanner;
2 class InvalidAgeException extends Exception{
3     InvalidAgeException(String msg){
4         super(msg);
5     }
6     public static void main(String args[]){
7
8         Scanner sc = new Scanner(System.in);
9         String name = sc.nextLine();
10        int age = sc.nextInt();
11        try{
12            if (age < 5 || age > 18)
13                throw new InvalidAgeException("Invalid age");
14
15            System.out.println("Student Name: " + name);
16            System.out.println("Student Age: " + age);
17            System.out.println("Registration Successful");
18
19        }catch (InvalidAgeException e) {
20            System.out.println(e.getMessage());
21        }
22    }
23 }
```

The "Input" field contains "arun" and "21". The "Output" field displays "Invalid age". The sidebar on the left shows two questions: "Q2 calculator application 10 marks" and "Q3 student registration system 10 marks", with "3/3 answered" indicated below.

